

Git Commands Cheat Sheet with Branching Strategy

1. Git Basics

Command	Use Case	Example
git init	Initialize a new git repository	git init my-project
git clone <repo_url>	Copy remote repo locally	git clone https://github.com/user/repo.git
git status	Show status of working directory	git status
git help <command>	Show help for a git command	git help commit

2. Configurations

Command	Use Case	Example
git config --global user.name "Your Name"	Set username	git config --global user.name "Pradeep Singh"
git config --global user.email "email@example.com"	Set email	git config --global user.email "pradeep@example.com"
git config --list	Show current configs	git config --list

3. Working with Files

Command	Use Case	Example
git add <file>	Stage file for commit	git add index.html
git add .	Stage all files	git add .
git reset <file>	Unstage a file	git reset index.html
git rm <file>	Remove file from repo	git rm old.txt
git mv <old> <new>	Rename/move file	git mv a.txt b.txt

4. Commits

Command	Use Case	Example
git commit -m "Message"	Commit staged changes	git commit -m "Added login feature"
git commit -am "Message"	Add & commit tracked files	git commit -am "Fixed bug in API"
git log	Show commit history	git log --oneline
git show <commit_id>	Show details of a commit	git show a1b2c3d

5. Branching

Command	Use Case	Example
git branch	List branches	git branch
git branch <name>	Create new branch	git branch feature/login
git checkout <branch>	Switch branch	git checkout main
git checkout -b <name>	Create & switch branch	git checkout -b hotfix
git merge <branch>	Merge branch into current	git merge feature/login
git branch -d <branch>	Delete branch	git branch -d old-feature

6. Remote Repositories

Command	Use Case	Example
git remote -v	List remotes	git remote -v
git remote add origin <url>	Add remote repo	git remote add origin https://github.com/user/repo.git
git push origin <branch>	Push branch to remote	git push origin main
git fetch	Fetch changes (no merge)	git fetch origin
git pull	Fetch + merge	git pull origin main

7. Undoing Changes

Command	Use Case	Example
git checkout -- <file>	Discard local changes	git checkout -- app.js
git reset --soft <commit>	Roll back, keep staged changes	git reset --soft HEAD~1
git reset --hard <commit>	Roll back, discard changes	git reset --hard HEAD~2
git revert <commit>	Create new commit that undoes a commit	git revert a1b2c3d

8. Stashing

Command	Use Case	Example
git stash	Save uncommitted changes	git stash
git stash pop	Apply last stash & remove	git stash pop
git stash list	Show stashes	git stash list

9. Collaboration & Review

Command	Use Case	Example
git diff	Show changes	git diff
git diff --staged	Show staged changes	git diff --staged
git blame <file>	Show who last changed each line	git blame index.html

10. Advanced

Command	Use Case	Example
git rebase <branch>	Reapply commits on top of another branch	git rebase main
git cherry-pick <commit>	Apply specific commit to current branch	git cherry-pick a1b2c3d
git tag <name>	Tag a version	git tag v1.0.0
git reflog	Show history of HEAD changes	git reflog

Git Branching Strategy

There are multiple branching strategies in Git, depending on project needs:

1. **Feature Branching**

- Each new feature gets its own branch off `main` or `develop`.
- Merged back once complete.
- Example: `git checkout -b feature/login`

2. **Git Flow**

- Long-lived `main` and `develop` branches.
- Features branch off `develop`.
- Release branches created for stabilization.
- Hotfixes branch off `main` for urgent fixes.

3. **GitHub Flow**

- Simpler workflow.
- Create a branch for each feature/fix from `main`.
- Open a Pull Request (PR).
- Merge once reviewed & tested.

4. **Trunk-Based Development**

- Developers commit small changes frequently to `main`.
- Short-lived branches (hours or a day).
- Relies heavily on CI/CD pipelines.

Best Practices:

- Keep branches small and focused.
- Always write descriptive commit messages.
- Regularly sync with remote (`git fetch` & `git pull`).
- Use Pull Requests for code reviews.